AWS STUDENT COMMUNITY DAY KAMPALA · BEGINNER TRACK

Your first homelab.

A beginner's guide to DevOps and Data Engineering. No experience needed.



ABOUT ME

Jacob Kipsang · Cloud Infrastructure Engineer

I design secure, scalable and cost-efficient cloud solutions. I have over a decade of hands-on experience in IT and cybersecurity, and I focus on driving innovation and operational excellence.

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```
$ docker run hello-world
```

The next 30 minutes

00–12 min

The basics

What a homelab is, key words, DevOps, data engineering and Docker.

12–24 min

Live demo

Run containers, put a web page online, store data and chart it.

24–30 min

Next steps

Homelab to AWS, a 7-day plan, free resources, and your questions.

WHAT IS A HOMELAB?

A computer you practise on, like a science lab, but for IT.

You try real tech skills on it, make mistakes safely, and learn by doing instead of only reading.

Six words you'll hear today

Server

A computer that stays on so people or programs can use it.

Linux

A free operating system, like Windows. Most servers run it.

Terminal

A window where you type commands instead of clicking.

Container

A sealed box with an app and everything it needs to run.

Database

An organised store of data, like a very smart spreadsheet.

Cloud

Someone else's servers you rent, like AWS. A homelab is your mini cloud.

Two jobs, explained simply

DEVOPS ENGINEER

Gets apps to users and keeps them running.

Think of a restaurant. The chef makes the food. DevOps makes sure every plate reaches the right table, hot, every single time.

DATA ENGINEER

Moves and cleans data so others can use it.

Think of plumbing. They lay the pipes so clean water, the data, flows to every tap where people need it.

DATA ENGINEERING

ETL: the heart of data engineering

ETL PIPELINE: EXTRACT → TRANSFORM → LOAD

E · EXTRACT

Collect

Pull data from forms, apps and files.

→

T · TRANSFORM

Clean

Fix mistakes, fill gaps, remove duplicates.

→

L · LOAD

Store

Save it in a database like PostgreSQL.

→

THEN

Show

Turn it into charts people can use.

Example: extract exam marks from a CSV → transform: fix typos and blank scores → load into PostgreSQL → show a bar chart in Metabase.

What you need (you probably have it)

01

A computer

Any laptop with 4 GB of RAM or more. An old one is perfect.

02

Free software

Ubuntu Linux, Docker and PostgreSQL. All free to download.

03

A little time

About one hour a day for one week.

On Windows 11? Use WSL: run **wsl --install** to get Ubuntu inside Windows, then install Docker Desktop.
On Windows 10 or older? Install VirtualBox and run Ubuntu inside it, then install Docker there.

WHAT IS DOCKER?

Docker is a shipping container for software.

Pack an app once, and it runs the same on your laptop, a friend's laptop, or a server on AWS.

LIVE DEMO

From zero to a website, a database and a chart, in five steps.

1 · DEVOPS

Run your first containers

2 · DEVOPS

Put your own web page online

3 · DATA

Start a database

4 · DATA

Store data and ask it questions

5 · DATA

Chart it with Metabase

LIVE DEMO · STEP 1 OF 5

Run your first containers

```
# compose.yml: step 1 is already on

$ docker compose up -d

$ docker compose logs hello

$ docker compose ps
```

WHAT YOU'LL SEE

"Hello from Docker!" in the logs.

Open **localhost:8080**: a web server is running.

ps lists your containers.

LIVE DEMO · STEP 2 OF 5

Put your own web page online

```
# compose.yml: uncomment
mysite:
  image: nginx
  ports:
    - "8081:80"
  volumes:
    - ./site:/usr/share/nginx/html

# terminal
$ docker compose up -d
```

WHAT YOU'LL SEE

Open **localhost:8081**: your page from **site/index.html**.

Edit the file, refresh, and it changes instantly. DevOps in miniature.

LIVE DEMO · STEP 3 OF 5

Start a database

```
# compose.yml: uncomment db + volumes at the bottom
db:
  image: postgres:16
  environment:
    POSTGRES_PASSWORD: pass
  ports:
    - "5432:5432"
  volumes:
    - pgdata:/var/lib/postgresql/data
# terminal
$ docker compose up -d
$ docker compose exec db psql -U postgres
```

WHAT YOU'LL SEE

A full PostgreSQL database starts in seconds.

The prompt changes to **postgres=#**: you're inside it.

LIVE DEMO · STEP 4 OF 5

Store data and ask it questions

```
CREATE TABLE marks (name TEXT, course TEXT, score INT);

INSERT INTO marks VALUES
    ('Amina','Cloud',78), ('Brian','Cloud',65),
    ('Grace','Data',88), ('Peter','Data',72);

SELECT course, ROUND(AVG(score), 1) AS avg
FROM marks GROUP BY course;
```

WHAT YOU'LL SEE

course	avg
Cloud	71.5
Data	80.0

Collect, store, ask questions: the heart of data engineering.

LIVE DEMO · STEP 5 OF 5

Chart it with Metabase

```
# compose.yml: uncomment
metabase:
  image: metabase/metabase
  ports:
    - "3000:3000"
  depends_on:
    - db

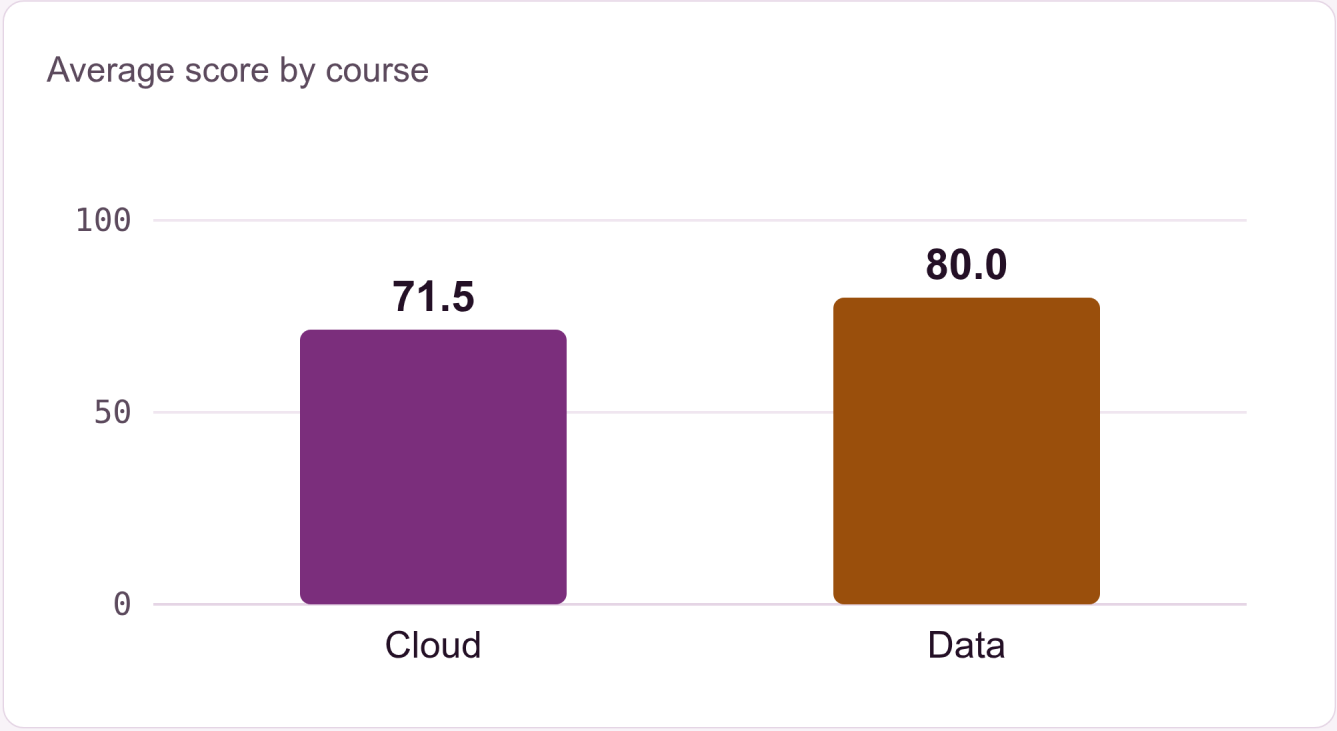
# terminal
$ docker compose up -d
```

WHAT YOU'LL SEE

1. Open **localhost:3000**
2. Add PostgreSQL: host **db**, user **postgres**, password **pass**
3. New → Question → **marks**
4. Average of score by course
5. Visualize → **Bar**

THE RESULT

Raw marks became a chart



Extract → Transform → Load → Visualize

Four containers, a few commands, and you've built a tiny version of what data engineers do every day.

HOMELAB → AWS

Same ideas, on AWS

DevOps & infrastructure

Homelab	On AWS
Laptop + Ubuntu	Amazon EC2
Docker + compose.yml	Amazon ECS + Fargate
Docker images	Amazon ECR
nginx + site/	Amazon S3 + CloudFront
pgdata volume	Amazon EBS
compose logs / ps	Amazon CloudWatch
Compose network	Amazon VPC

Data engineering

Homelab	On AWS
CSV / raw files	Amazon S3 (data lake)
ETL pipeline (Apache NiFi, Airbyte, dbt)	AWS Glue
PostgreSQL (db)	Amazon RDS
Big data warehouse (ClickHouse, Apache Hive)	Amazon Redshift
SQL on files	Amazon Athena
cron / Apache Airflow	Amazon MWAA
Metabase	Amazon Quick Sight

Your 7-day beginner plan

DAY 1

Set up Linux

Windows 11: run wsl --install. Windows 10 or older: Ubuntu in VirtualBox.

DAY 2

Learn the terminal

Practise ls, cd, mkdir, cp and nano until they feel normal.

DAY 3

Install Docker

Run hello-world. Your first container!

DAY 4 · DEVOPS

Run a website

Start nginx and change its home page to say your name.

DAY 5 · DATA

Run a database

Start PostgreSQL in Docker and create your first table.

DAY 6 · DATA

Make a chart

Load a CSV of your choice and chart it in Metabase.

DAY 7

Share it

Write what you learned and put your notes on GitHub.

AFTER THAT

Level up

Try automatic deployments, monitoring and data pipelines.

Four beginner rules

Breaking it is normal

If something breaks, you're learning. You can always reinstall and start again.

Search the error

Copy the error message into Google or an AI assistant. Someone has seen it before.

Write it down

Keep every command you run in a notes file. It becomes your portfolio later.

Keep it private

Don't open your homelab to the internet yet. Learn on your own network first.

Free places to learn

Topic	Resource	Where
Linux	Linux Journey (short, friendly lessons)	linuxjourney.com
Docker	Official “Get started” guide	docs.docker.com
Networking	Open textbook: An Introduction to Computer Networks	intronetworks.cs.luc.edu
Docker networks	How containers talk to each other	docs.docker.com/engine/network
Cloud	AWS Cloud Practitioner Essentials	skillbuilder.aws
What next	Step-by-step learning maps	roadmap.sh
People	Learn together with your peers	AWS communities

START TONIGHT

Just start.

One laptop, one hour, one command: docker run hello-world.

Questions?

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